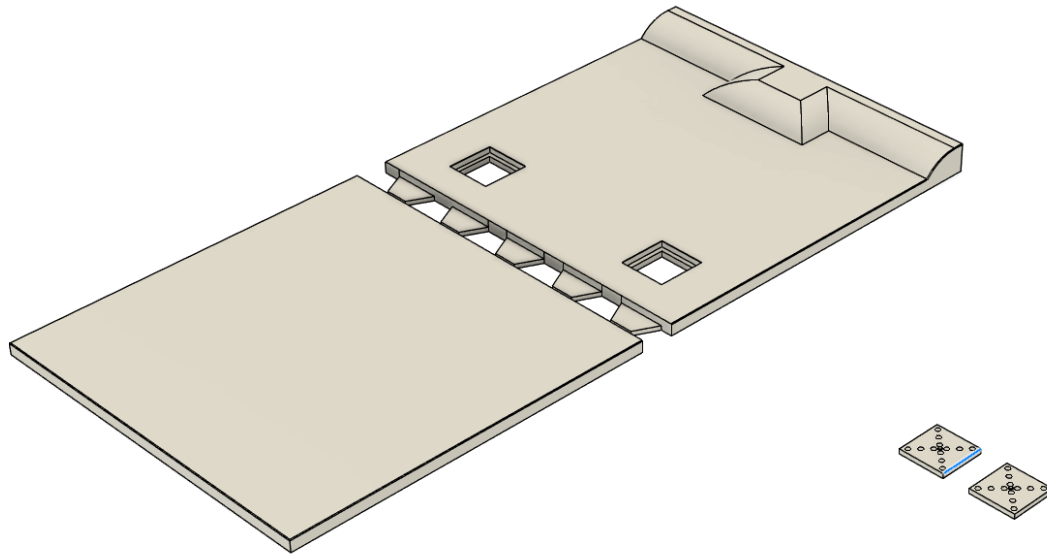


Project C2: Capillary Canopy

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Technical Field:

This project involves the technical fields of additive manufacturing (FDM 3D printing), CAD product design (Autodesk Fusion 360), microclimate / humidity control in sealed enclosures, and fluid behavior in small systems (evaporation, condensation, capillary wicking)

Background Information:

The Capillary Canopy concept came from a very normal hobbyist experience: small tanks are either *open and annoying* (evaporation, dust, constant top-offs) or *covered but dumb* (generic lids that don't manage condensation and don't support a stable internal humidity cycle). The idea is to design and 3D-print (on a Bambu P1P) a set of purpose-built lids for three 2.5-gallon glass tanks, each lid tuned to create a repeatable “mini climate” using geometry—condensation channels, drip rails, optional vents, and a reservoir-fed wick tube—rather than relying on external equipment. All modeling and iteration

happen in Fusion 360, so the whole project lives cleanly in a CAD → prototype → test loop.

Prior Art (Research):



There are barely any options for 2.5-gallon open glass aquariums. There is a solid glass or a metal screen option, basically totally closed or fully ventilated.

My prototype is different because it will provide functionality to the tank instead of just limiting what can enter or exit it. The lid sustains the environment of the tank instead of vice versa.

Project Description:

This project designs and prototypes three 3D-printed lids for three 2.5-gallon glass tanks, modeled in Fusion 360 and printed on a Bambu P1P. Each lid will be a tuned variant (ex: arid / temperate / humid behavior) using the same core architecture: a sealed rim interface, controlled vent options, and a reservoir-fed wick tube that supports passive moisture movement. The lids will incorporate surface features—like internal slopes, drip rails, and condensation channels—to intentionally route condensed water (ideally away from the front viewing pane) and reduce user maintenance. The intended users are hobbyists, students, and “I want a tiny ecosystem, but I also want

my weekends back” people who want a stable micro-environment without external pumps or constant attention.

Innovation Claim:

This project is innovative because it uses 3D-printed lid geometry. Not external equipment to create repeatable, tunable microclimates in small tanks, combining condensation routing, optional vent control, and an integrated reservoir + wick tube into a single modular lid system manufactured on a Bambu P1P from Fusion 360 models.

Usage Scenario:

You set up each 2.5-gallon tank with substrate/scape appropriate to the target humidity profile, install the printed lid, fill the lid reservoir, and let the system passively cycle moisture (evaporation → condensation → guided return). Periodically you top off the reservoir (frequency depends on the variant and ambient room conditions), and you can tweak behavior by swapping vent inserts or switching to a different lid variant.

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Evaluation Criteria:

- Does the lid fit securely within the 11.0625" × 4.9375" inner lip without excessive play + or - .03.
- Does the lid remain structurally rigid under normal handling without visible flexing?
- Does the lid rest evenly on the black rim without rocking?
- Does the lid remain seated after 30 days without warping?
- Does the system operate without external pumps, misters, or electronics?
- Does the lid prevent insect entry through openings larger than 2 mm?
- Does the lid maintain structural integrity in high-humidity conditions (over 80%)?
- Can the lid be removed and reinstalled without damaging the tank rim?
- Are all components manufacturable using Fusion 360?
- Can wick inserts be removed and replaced without damaging the printed tube?
- Are all internal channels free of stagnant water accumulation after 7 days?
- Does the Arid lid reduce soil moisture retention compared to the Tropical lid?

Goals and Tasks Associated with the Project:

Goal 1: Develop Biome-Specific Passive Lid Designs:

Design three distinct lid geometries (Arid, Temperate, Tropical) that modify internal condensation behavior without altering tank water volume.

Goal 1: Tasks – Biome Lid Design

- Measure tank outer and inner lip dimensions.
- Create parametric base model in Fusion 360.
- Model Arid interior slope geometry.
- Model Temperate dome and perimeter channel.
- Model Tropical ribbing and deep basin.
- Define channel slope angles (1–5°).
- Model return tube placement and diameter.
- Perform section analysis to verify water paths.
- Adjust interior geometries for print feasibility.

Goal 2: Engineer a Secure and Repeatable Fit:

Create a dual-flange lid structure that reliably interfaces with the standard 2.5-gallon Aqueon or other basic open glass black rimmed aquarium.

Goal 2: Tasks – Fit Engineering

- Design upper support flange.
- Design lower insert flange.
- Define tolerance (0.015"–0.025").
- Prototype tolerance test print ring.
- Adjust CAD dimensions based on physical test.
- Verify snap-in seating stability.
- Test lid removal and reinsertion durability.

Goal 3: Prototype and Manufacture Using Additive Fabrication

Produce functional physical prototypes using Fusion 360 and a Bambu P1P printer.

Goal 3: Tasks – Fabrication

- Slice models in Bambu Studio.
- Optimize orientation for minimal support.
- Define wall thickness and infill.
- Print Arid prototype.
- Print Temperate prototype.
- Print Tropical prototype.
- Post-process and clean internal channels.
- Install wick inserts (Tropical).

Goal 4: Validate Performance with Controlled Testing

Measure humidity, condensation behavior, and soil moisture stability over a 4-week test period.

Goal 4: Tasks – Testing & Data Collection

- Set up three identical tanks. (Possibly 4, with 2 Tropical; 1 wick vs 2 wick)
- Use identical substrate and water volume.
- Place tanks in same environmental conditions.
- Measure daily humidity.
- Record condensation patterns.
- Observe soil moisture levels.
- Document glass clarity.
- Track wick flow behavior. (Tropical)
- Record 4-week data.
- Compare results between designs.

Goal 5: Demonstrate Environmental Control Through Geometry

Prove that climate variation can be achieved solely through structural design without mechanical or electrical systems.

Goal 5: Tasks – Validation & Documentation

- Photograph prototypes.
- Create CAD renders.
- Compile humidity comparison table.
- Analyze differences between lid types.
- Record conclusions.
- Prepare presentation slides.
- Delivery of final presentation.

Description of Design Prototype:

Through evaporation, moisture rises through vents in the internal lid components, circulating back into the lid itself, it then pools and returns to the tank through specific channels that guide it.

Evaluation Plan

The Capillary Canopy lids will be evaluated through controlled, side-by-side testing of three biome variants (Arid, Temperate, Tropical) over a 4-week period.

Three identical 2.5-gallon tanks will be set up using the same substrate, water volume, and environmental placement to eliminate external variability. Each tank will receive a different lid variant.

Data collection will include:

- Humidity Tracking:
Daily humidity readings (same time each day) to compare stability and variance across designs.
- Condensation Behavior:
Visual documentation (photos) of condensation patterns, specifically:

- Formation points
 - Flow direction
 - Effectiveness of channel routing
 - Presence of front-glass obstruction
- Soil Moisture Retention:
Observational comparison of substrate dampness over time to validate biome differentiation (Arid vs Tropical performance).
- Wick Performance (Tropical Variant):
Monitoring capillary movement efficiency:
 - Consistency of moisture delivery
 - Signs of over-saturation or dry-out
- Evaporation / Reservoir Loss:
Measuring how frequently the reservoir requires refilling as an indicator of system efficiency.
- Structural Performance:
Weekly inspection for:
 - Warping
 - Fit degradation
 - Material performance in high humidity

At the end of testing, results will be compiled into a comparison table to determine whether geometry alone successfully creates distinct and stable microclimates.

Project Completion Assessment

The project will be considered complete when the following conditions are met:

- All three lid variants (Arid, Temperate, Tropical) are successfully designed, printed, and installed on test tanks.
- Each lid demonstrates a measurable and repeatable difference in environmental behavior, specifically:
 - Lower humidity and faster dry-out in Arid
 - Balanced moisture retention in Temperate

- High humidity and sustained moisture in Tropical
- Condensation is actively controlled and routed, with:
 - Minimal obstruction of the front viewing pane
 - No uncontrolled dripping or pooling
- The system operates fully passively, with:
 - No pumps, electronics, or external intervention
 - Only periodic reservoir refilling required
- All lids maintain structural integrity and fit over the full 4-week test period.
- Documentation is complete, including:
 - Photos of prototypes and test setups
 - CAD renders from Fusion 360
 - Data tables and comparison results
 - Final presentation materials

If these conditions are met, the project successfully proves the core innovation claim: that geometry alone can control environmental behavior in a closed system.

The wick's became moldy in a way that even the insects couldn't clean them, they require to much maintenance to be viable and the system works good if not the same without. They actually require less water in the system without the wick.

Ultimately the lid did work as intended, but without an outside source of heat temperate seems to be the only sustainable option.

Appendices

The following materials will be included as supporting documentation:

- **Appendix A: CAD Models (Fusion 360)**
- **Appendix B: Prototype Images**
- **Appendix B: Final Images**